

1. Children of three ages are asked to indicate their preference for three photographs of adults.

Do the data suggest that there is a significant relationship between age and photograph preference? What is wrong with this study? (10 Marks)

Photograph:

Age of child A B C

5-6 years: 18 22 20

7-8 years: 2 28 40

9-10 years: 20 10 40

Use `cov()` to calculate the sample covariance between B and C.

Use another call to `cov()` to calculate the sample covariance matrix for the preferences.

Use `cor()` to calculate the sample correlation between B and C.

Use another call to `cor()` to calculate the sample correlation matrix for the preferences.

2. Assume the Tennis coach wants to determine if any of his team players are scoring outliers.

To visualize the distribution of points scored by his players, then how can he decide to develop the box plot? (10 Marks)

Player

ID

Player

Name

Points

Scored

P1 Player A 12

P2 Player B 15

P3 Player C 14

P4 Player D 10

P5 Player E 18

P6 Player F 20

P7 Player G 22

P8 Player H 13

P9 Player I 11

P10 Player J 35

P11 Player K 16

P12 Player L 17

P13 Player M 19

P14 Player N 21

P15 Player O 23

3. A bank wants to decide whether to approve a loan based on customer attributes like:

☐ Age

☐ Income

☐ Employment Status

☐ Credit Score

Using historical data, build a Decision Tree classifier to predict Loan Approval (Yes/No).

(10 Marks)

Consider the Data Set

young,high,employed,good,yes

young,high,self-employed,average,yes

middle,high,employed,good,yes

old,medium,employed,good,yes

old,low,unemployed,poor,no

old,low,self-employed,average,no

middle,low,unemployed,poor,no

young,medium,employed,average,yes

young,low,unemployed,poor,no

old,medium,self-employed,good,yes

middle,medium,employed,average,yes

middle,high,self-employed,good,yes

old,medium,unemployed,poor,no

young,medium,self-employed,average,yes

middle,low,employed,poor,no

4. Two Maths teachers are comparing how their Year 9 classes performed in the end of year

exams. Their results are as follows: (10 Marks)

Class A: 76, 35, 47, 64, 95, 66, 89, 36, 8476,35,47,64,95,66,89,36,84

Class B: 51, 56, 84, 60, 59, 70, 63, 66, 5051,56,84,60,59,70,63,66,50

(i) Find which class had scored higher mean, median and range.

(ii) Plot above in boxplot and give the inferences

5. Suppose that a hospital tested the age and body fat data for 18 randomly selected adults

with the following results: (10 Marks)

(a) Calculate the mean, median, and standard deviation of age and %fat.

(b) Draw the boxplots for age and %fat.

(c) Draw a scatter plot and a q-q plot based on these two variables.

<i>age</i>	23	23	27	27	39	41	47	49	50
<i>%fat</i>	9.5	26.5	7.8	17.8	31.4	25.9	27.4	27.2	31.2
<i>age</i>	52	54	54	56	57	58	58	60	61
<i>%fat</i>	34.6	42.5	28.8	33.4	30.2	34.1	32.9	41.2	35.7

Absolutely. Below is a **complete 10-mark style answer** for all 5 questions, including the **R commands** using `cov()`, `cor()`, boxplots, decision tree, scatter plot and Q-Q plot.

1. Age and Photograph Preference

Given data

Age	A	B	C	Total
-----	-:	-:	-:	----
5–6 years	18	22	20	60
7–8 years	2	28	40	70
9–10 years	20	10	40	70

(a) Test for significant relationship

We use a **Chi-square test of independence** because both variables are categorical:

* Variable 1: Age group

* Variable 2: Photograph preference

Hypotheses

* **H_0** : Age and photograph preference are independent.

* **H_1** : Age and photograph preference are significantly related.

R code

```
```r
```

```
data <- matrix(c(18,22,20,
 2,28,40,
 20,10,40),
 nrow = 3, byrow = TRUE)
```

```
colnames(data) <- c("A", "B", "C")
```

```
rownames(data) <- c("5-6", "7-8", "9-10")
```

```
data
```

```
chisq.test(data)
```

```
```
```

Result

The calculated Chi-square statistic is approximately:

```
[  
 \chi^2 = 29.603  
]
```

Degrees of freedom:

```
[  
 df=(3-1)(3-1)=4
```

]

p-value:

[

$p \approx 0.0000059$

]

Since:

[

$p < 0.05$

]

we **reject H_0** .

Conclusion

****There is a statistically significant relationship between the age of the child and photograph preference.****

(b) What is wrong with this study?

The major limitation is that the study does ****not control for other factors that may influence photograph preference****.

For example:

- * The photographs may differ in attractiveness, facial expression, gender, etc.
- * The order in which photographs were shown may influence preference.
- * The study does not clearly state whether the children were randomly selected.
- * Therefore, the study can establish an **association**, but it cannot establish that age itself **causes** the difference in preference.

(c) Sample covariance between B and C

B:

```
[
B=(22,28,10)
]
```

C:

```
[
C=(20,40,40)
]
```

R command

```
` ` `r
```

```
B <- c(22, 28, 10)
```

```
C <- c(20, 40, 40)
```

```
cov(B, C)
```

```
`, `
```

```
### Answer
```

```
[
```

```
\boxed{\operatorname{Cov}(B,C)=-20}
```

```
]
```

```
---
```

```
## (d) Sample covariance matrix
```

```
` ``r
```

```
preferences <- data.frame(
```

```
  A = c(18,2,20),
```

```
  B = c(22,28,10),
```

```
  C = c(20,40,40)
```

```
)
```

```
cov(preferences)
```

```
`, `
```

```
### Covariance matrix
```

```
[
```

```
\boxed{
```



```

\begin{bmatrix}
97.33 & -74.00 & -46.67 \\
-74.00 & 84.00 & -20.00 \\
-46.67 & -20.00 & 133.33
\end{bmatrix}

```

(e) Sample correlation between B and C

```

```r

```

```

cor(B, C)

```

```

```

```

Answer

```

[
\boxed{r_{BC}=-0.18898}
]

```

This indicates a **weak negative correlation** between B and C.

(f) Sample correlation matrix

```
```\r
```

```
cor(preferences)
```

```
```\r
```

```
### Answer
```

```
[
```

```
\boxed{
```

```
\begin{bmatrix}
```

```
1 & -0.8184 & -0.4096\
```

```
-0.8184 & 1 & -0.1890\
```

```
-0.4096 & -0.1890 & 1
```

```
\end{bmatrix}
```

```
}
```

```
]
```

```
---
```

2. Tennis Players – Identifying Scoring Outliers

```
### Data
```

```
| Player | Points |
```

```
| ----- | -----: |
```

```
| A   | 12 |
```

```
| B   | 15 |
```

```
| C   | 14 |
```

```
| D   | 10 |
```

E	18
F	20
G	22
H	13
I	11
J	**35**
K	16
L	17
M	19
N	21
O	23

R code

```
```r
```

```
points <- c(12,15,14,10,18,20,22,13,11,35,
 16,17,19,21,23)
```

```
boxplot(points,
 main = "Tennis Players' Points",
 ylab = "Points Scored",
 col = "lightblue")
```

```
```
```

Five-number summary

```
```r
```

```
summary(points)
```

...

Important values:

$$* Q1 = **13.5**$$

$$* \text{Median} = **17**$$

$$* Q3 = **20.5**$$

Therefore:

$$\begin{aligned} &[ \\ &\text{IQR} = Q3 - Q1 \\ &] \end{aligned}$$

$$\begin{aligned} &[ \\ &\text{IQR} = 20.5 - 13.5 = 7 \\ &] \end{aligned}$$

### Lower fence

$$\begin{aligned} &[ \\ &13.5 - 1.5(7) = 3 \\ &] \end{aligned}$$

### Upper fence

$$\begin{aligned} &[ \\ &20.5 + 1.5(7) = 31 \end{aligned}$$

]

Any value above 31 is an outlier.

Since:

[

$35 > 31$

]

**\*\*Player J, who scored 35 points, is an outlier.\*\***

### Inference

The boxplot will show most players' scores concentrated between approximately **\*\*10 and 23\*\***, while **\*\*35** appears as an isolated point**\*\***.

[

`\boxed{\text{Player J is a scoring outlier}}`

]

---

# 3. Bank Loan Approval – Decision Tree Classifier

We have four input attributes:

\* Age

- \* Income
- \* Employment Status
- \* Credit Score

Target:

- \* Loan Approval: Yes/No

---

## R implementation

### Step 1: Create the dataset

```
` `` `r
```

```
data <- data.frame(
```

```
 Age = c("young","young","middle","old","old",
 "old","middle","young","young","old",
 "middle","middle","old","young","middle"),
```

```
 Income = c("high","high","high","medium","low",
 "low","low","medium","low","medium",
 "medium","high","medium","medium","low"),
```

```
 Employment = c("employed","self-employed","employed",
 "employed","unemployed","self-employed",
 "unemployed","employed","unemployed",
 "self-employed","employed","self-employed",
```

```
"unemployed","self-employed","employed"),
```

```
Credit = c("good","average","good","good","poor",
 "average","poor","average","poor","good",
 "average","good","poor","average","poor"),
```

```
Loan = c("yes","yes","yes","yes","no",
 "no","no","yes","no","yes",
 "yes","yes","no","yes","no")
```

```
)
```

```
data
```

```
```\n
```

```
---
```

```
## Step 2: Convert variables into factors
```

```
```\n
```

```
data[] <- lapply(data, factor)
```

```
```\n
```

```
---
```

```
## Step 3: Install and load `rpart`
```

```
```\n
```

```
install.packages("rpart")
```

```
install.packages("rpart.plot")
```

```
library(rpart)
```

```
library(rpart.plot)
```

```
```\n
```

```
---
```

```
## Step 4: Build the decision tree
```

```
```\n
```

```
model <- rpart(
```

```
 Loan ~ Age + Income + Employment + Credit,
```

```
 data = data,
```

```
 method = "class"
```

```
)
```

```
model
```

```
```\n
```

```
---
```

```
## Step 5: Plot the tree
```

```
```\n
```

```
rpart.plot(model,
```

```
 main = "Decision Tree for Loan Approval")
```

```
```\n
```


Logical structure of the tree

The strongest first splitting variable is **Credit Score**.

```text

```
 Credit Score
 / | \
Good Average Poor
 | | |
YES Age NO
 / \
 Old Young/Middle
 | |
 NO YES
```

```

Interpretation

* **Good credit score** → Loan Approved*

* **Poor credit score** → Loan Rejected*

* **Average credit score** → Age is considered*

* Old → No

* Young/Middle → Yes

Conclusion

The decision tree can use historical customer information to classify whether a new customer's loan should be ****approved or rejected****.

4. Comparing Maths Classes

Class A

```text

76, 35, 47, 64, 95, 66, 89, 36, 84

```

Class B

```text

51, 56, 84, 60, 59, 70, 63, 66, 50

```

(i) Mean, median and range

R code

```r

```
classA <- c(76,35,47,64,95,66,89,36,84)
```

```
classB <- c(51,56,84,60,59,70,63,66,50)
```

```
mean(classA)
median(classA)
range(classA)
` ``
```

```
` ``r
mean(classB)
median(classB)
range(classB)
` ``
```

### ### Results

| Measure | Class A          | Class B |
|---------|------------------|---------|
| -----   | -----:           | -----:  |
| Mean    | <b>**65.78**</b> | 62.11   |
| Median  | <b>**66**</b>    | 60      |
| Minimum | 35               | 50      |
| Maximum | 95               | 84      |
| Range   | <b>**60**</b>    | 34      |

### ### Conclusion

Class A has:

- \* Higher mean
- \* Higher median

\* Larger range

Therefore, **\*\*Class A performed better on average\*\***, but its marks were also much more spread out.

---

# Boxplot

### R code

```
` `` `r
```

```
scores <- list(
 "Class A" = classA,
 "Class B" = classB
)
```

```
boxplot(scores,
 main = "Comparison of Year 9 Class Scores",
 ylab = "Exam Scores",
 col = c("lightblue", "lightgreen"))
` `` `
```

### Inferences from the boxplot

1. **\*\*Class A has a higher median\*\*** than Class B.
2. Class A has **\*\*greater variability\*\*** in scores.
3. Class A has a much larger range.

4. Class B's scores are more concentrated.
5. Class A contains both relatively low and very high scores.
6. Class A has the overall higher average performance.

### Final conclusion

[  
\boxed{\text{Class A performed better, but Class A had greater variability.}}  
]

---

#### # 5. Hospital Age and Body-Fat Data

From the image, the 18 observations are:

| Age   %Fat |
|------------|
| --:   ---: |
| 23   9.5   |
| 23   26.5  |
| 27   7.8   |
| 27   17.8  |
| 39   31.4  |
| 41   25.9  |
| 47   27.4  |
| 49   27.2  |
| 50   31.2  |
| 52   34.6  |

| 54 | 42.5 |

| 54 | 28.8 |

| 56 | 33.4 |

| 57 | 30.2 |

| 58 | 34.1 |

| 58 | 32.9 |

| 60 | 41.2 |

| 61 | 35.7 |

---

## R data entry

```r

```
age <- c(23,23,27,27,39,41,47,49,50,  
        52,54,54,56,57,58,58,60,61)
```

```
fat <- c(9.5,26.5,7.8,17.8,31.4,25.9,  
        27.4,27.2,31.2,34.6,42.5,28.8,  
        33.4,30.2,34.1,32.9,41.2,35.7)
```

```

---

# (a) Mean, Median and Standard Deviation

## Age

```
```\r
mean(age)
median(age)
sd(age)
```\r
```

Results:

```
[
\boxed{\text{Mean Age}=46.44}
]
```

```
[
\boxed{\text{Median Age}=51}
]
```

```
[
\boxed{\text{SD}=13.22}
]
```

---

## %Fat

```
```\r
mean(fat)
median(fat)
sd(fat)
```

...

Results:

```
[  
  \boxed{\text{Mean Fat}=28.78}  
]
```

```
[  
  \boxed{\text{Median Fat}=30.70}  
]
```

```
[  
  \boxed{\text{SD}=9.25}  
]
```

Summary

Variable	Mean	Median	Standard Deviation
-----	-----:	-----:	-----:
Age	**46.44**	**51.00**	**13.22**
%Fat	**28.78**	**30.70**	**9.25**

> `sd()` in R gives the **sample standard deviation**, which is appropriate here for randomly selected adults.

(b) Boxplots

Separate boxplots

```
` ``r
par(mfrow=c(1,2))

boxplot(age,
        main="Boxplot of Age",
        ylab="Age")

boxplot(fat,
        main="Boxplot of Body Fat %",
        ylab="% Fat")

par(mfrow=c(1,1))
` ``
```

Or a combined boxplot:

```
` ``r
boxplot(age, fat,
        names=c("Age", "% Fat"),
        main="Boxplots of Age and Body Fat")
` ``
```

Five-number summaries

```
` ``r
```

```
summary(age)
```

```
summary(fat)
```

```
` ``
```

For **Age**:

* Minimum = 23

* Q1 = 39.5

* Median = 51

* Q3 = 56.75

* Maximum = 61

For **%Fat**:

* Minimum = 7.8

* Q1 = 26.675

* Median = 30.7

* Q3 = 33.925

* Maximum = 42.5

There are **no strong boxplot outliers** using the usual $1.5 \times \text{IQR}$ rule.

(c) Scatter Plot

A scatter plot can be used to examine the relationship between age and body fat percentage.

```
```r
plot(age, fat,
 main="Age vs Body Fat %",
 xlab="Age",
 ylab="Body Fat %",
 pch=19)
```

```
abline(lm(fat ~ age))
```
```

Correlation

```
```r
cor(age, fat)
```
```

Result:

```
[
\boxed{r=0.8176}
]
```

This indicates a **strong positive relationship** between age and body fat percentage.

As age increases, body fat percentage generally tends to increase.

The fitted regression equation is approximately:

```
[  
\boxed{\%Fat=2.198+0.5724(Age)}  
]
```

Q-Q Plot

If the question asks for a Q-Q plot comparing the two variables:

```
```r  
qqplot(age, fat,
 main="Q-Q Plot: Age vs Body Fat %",
 xlab="Age Quantiles",
 ylab="Body Fat % Quantiles")
```

```
abline(0, 1)
```

```
```
```

If the intention is to check **normality** of each variable, use:

```
```r  
par(mfrow=c(1,2))

qqnorm(age, main="Q-Q Plot of Age")
```

```
qqline(age)
```

```
qqnorm(fat, main="Q-Q Plot of Body Fat %")
```

```
qqline(fat)
```

```
par(mfrow=c(1,1))
```

```
```\n
```

```
### Interpretation
```

* Points close to the straight line indicate approximate normality.

* Deviations from the line indicate departures from normality.

* The scatter plot shows a **strong positive association** between age and body fat.

```
---\n
```

```
# Final Answers at a Glance
```

Question	Main Result	
-----	-----	

1	Significant relationship between age and photograph preference, $\chi^2 = 29.603$, $p < 0.001$	
1 Cov(B,C)	-20	
1 Cor(B,C)	-0.18898	
2	Player J (35 points) is an outlier	
3	Decision tree: Credit Score $\rightarrow$ Age for average credit	
4 Class A Mean	65.78	

4 Class B Mean	**62.11**	
4 Class A Median	**66**	
4 Class B Median	**60**	
4 Class A Range	**60**	
4 Class B Range	**34**	
5 Age Mean	**46.44**	
5 Age Median	**51**	
5 Age SD	**13.22**	
5 %Fat Mean	**28.78**	
5 %Fat Median	**30.70**	
5 %Fat SD	**9.25**	
5 Age–Fat correlation	**0.8176 (strong positive)**	